



SCS 1306 - Linear Algebra

Assignment 01

Duration: 1 Hour 45 Minutes

Answer all questions.

Question 01 (40 Marks)

Let

$$A(t) = \begin{bmatrix} 1 & 1 & 1 \\ 1 & t & 2 \\ 1 & 2 & t \end{bmatrix}, \quad b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}.$$

- (a) Find $\det(A(t))$ using cofactor expansion.
- (b) For which values of t does $A(t)x = b$ have a unique solution for every $b \in \mathbb{R}^3$?
- (c) When $t = 0$, use row reduction to find the condition on b_1, b_2, b_3 for the system to be consistent.
- (d) When $t = 2$, use row reduction to find the condition on b_1, b_2, b_3 for the system to be consistent.
- (e) Explain geometrically what happens to the column space when $t = 0$ or $t = 2$. Is the column space all of $\mathbb{R}^3$, a plane, a line, or only the zero vector?

Question 02 (30 Marks)

Let

$$n = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}, \quad a = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \quad b = \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}.$$

Consider the plane

$$P = \{x \in \mathbb{R}^3 : n \cdot x = 0\}.$$

- (a) Show that both a and b lie in the plane P .
- (b) Show that a and b are linearly independent.
- (c) Find the angle between a and b using the cosine formula

$$\cos \theta = \frac{a \cdot b}{\|a\| \|b\|}.$$

- (d) Express

$$r = \begin{bmatrix} 3 \\ 2 \\ -2 \end{bmatrix}$$

as a linear combination of a and b . Hence explain why $r \in P$.

- (e) Give a geometric explanation of why all vectors satisfying $n \cdot x = 0$ form a plane through the origin.

Question 03 (30 Marks)

Let

$$A = \begin{bmatrix} 2 & 4 & 6 \\ 1 & 2 & 3 \\ -1 & -2 & -3 \end{bmatrix}.$$

- (a) Show that all columns of A lie on the same line. Hence state the rank of A .
- (b) Describe the column space $C(A)$ geometrically.
- (c) Determine whether the system

$$Ax = \begin{bmatrix} 12 \\ 6 \\ -6 \end{bmatrix}$$

has no solution, one solution, or infinitely many solutions. Find all solutions if they exist.

- (d) Determine whether

$$Ax = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

is solvable. Justify your answer using the column space.

- (e) Use cofactor expansion to compute $\det(A)$. Explain why A is not invertible.